

Quantum recovery from finitely many sharp questions

Seth Douglas
seth.douglas@gmail.com

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Abstract

Suppose that after a quantum channel each observable in a finite family of sharp binary questions can be approximately recovered by its own output measurement. We study when these separate guarantees imply recovery of the entire quantum input, including its correlations with an arbitrary reference. A positive word-averaging certificate gives a linear diamond-error bound. For reflection families of fixed degree and fixed lazy Hilbert–Schmidt spectral gap, a word-filtration noise construction proves that the resulting $\Theta(\min\{1, e(\log d)^2\})$ minimax scaling is optimal. This certificate is not intrinsically optimal for every family: coordinate transpositions and one rank-one sign flip have recovery modulus $\Theta(\min\{1, de\})$, although the positive-word bound of the stated form necessarily pays order d^2e . We also give decoders determined by the supplied effects with linear half-diamond error uniformly over compatible channels: $O(de)$ for a rank-one-anchor family and $O(e/\sin^2 \theta)$ from any nonparallel qubit pair with angle θ . Finally, for $n = 3s$, seven explicitly specified test types admit an executable decoder using $O(n)$ reversible tester calls and delivering $O(n^2e)$ half-diamond error. The distinction between mathematical existence, channel-independent choice and coherent access is essential throughout.

1. Introduction

Recovering the answers to several classical questions about a quantum input is weaker, on its face, than recovering that input as a quantum system. Different questions may use incompatible physical measurements. Even when their ideal observables generate the full matrix algebra, approximate preservation need not propagate without quantitative loss.

This issue is already visible in approximate observable error correction. Bény emphasizes that the exact principle of checking algebra generators does not automatically provide a uniform approximate bound on their products [B09]. Complementary-observable recovery provides a constructive positive precedent: Renes obtains entanglement recovery from coherent realizations of classical guessing measurements [R16]. Our premise is a worst-input operator-norm calibration of finitely many sharp binary effects. Our output criterion is the full diamond norm, and we distinguish precisely what the decoder may know.

The first contribution is a quantitative implication from this premise to full recovery. It uses word averaging, complementary-channel fidelity and a positive replacer component. A matching lower bound explains why the dimension-dependent loss cannot be removed even for fixed-gap families. The second contribution separates this averaging certificate from the true recovery cost through an explicit signed-permutation family. Direct anchor extraction beats every bound of the stated positive-word form by a factor of order d . The third contribution gives linear uniform bounds for effects-determined anchor and qubit decoders, and an explicitly counted compiler from seven supplied coherent tester types.

These are successive parts of one problem. They do not establish a universal intrinsic recovery law for all finite question geometries, nor do they show that suitable physical testers can always be efficiently supplied.

1.1. Relation to earlier recovery criteria

The Heisenberg formulation is foundational rather than new here. Bény, Kempf and Kribs [BKK07, Theorem 9 and Corollary 10] characterize exactly correctable observable algebras through compressed Kraus-product commutants. Their framework includes quantum/classical direct sums and subsystems. Bény’s approximate extension [B09, Theorem 2] shows that environment commutator error at most η , uniformly over all norm-one algebra elements and amplified environmental operators, implies unhalved diamond reconstruction error at most $2\sqrt{\eta}$. In our half-diamond convention this is $\sqrt{\eta}$. Applying that criterion directly would still require deriving its whole-algebra premise from our finite tests; a linear bound through that estimate would require $\eta = O(e^2)$. Its warning about accumulating product errors motivates the modulus here. Our input consists instead of separately optimized output effects for finitely many logical projections. Passing quantitatively from that premise to a whole-algebra recovery guarantee is the task, not the introduction of observable error correction.

Renes [R10, Theorem 1] gives coherent entanglement recovery from complementary guessing measurements; [R16, Theorem 1, Lemma 1 and Eqs. (8)–(12)] supersedes that work and explicitly composes coherent realizations of those measurements. For a complementary qubit pair our calibration implies guessing probabilities at least $1 - e$. Applying that construction to the normalized Choi state gives root fidelity at least $1 - 4e + 2e^2$ in the nontrivial small-error regime. The normalized-Choi to diamond conversion then bounds the same effects-determined decoder’s half-diamond error by $\min\{1, 4\sqrt{2e}\}$. Thus channel-independent choice from fixed complementary measurements is already established. The additional content of Theorem 6.1 is the conversion from an arbitrary oblique irreducible qubit pair to a common exact Pauli representation with linear $O(e/s^2)$, hence $O(\Gamma_{\text{cb}}^2 e)$, worst-case diamond error. Theorem 7.2 separately supplies the specified multi-qubit tester compiler. These bounded comparisons do not exclude a sharper adaptation of earlier constructive proofs.

Nakata, Matsuura and Koashi [NMK25, Definition 1 and Theorem 1] also combine two classical decoding measurements into a quantum decoder, extending a CSS decoding circuit to general codes. For complementary bases, their basis-averaged classical errors at most e imply normalized-Choi trace-distance error at most $(1 + \sqrt{2})\sqrt{e}$ [NMK25, Corollary 1]. Their Eq. (12) converts this to at most $d(1 + \sqrt{2})\sqrt{e}$ in half-diamond distance with the same decoder. For a fixed noncomplementary qubit pair, the displayed bound retains a complementarity term even at zero classical error; it does not give the vanishing oblique-pair estimate here.

Kretschmann, Schlingemann and Werner [KSW06] and Bény–Oreshkov [BO10, Theorem 1] provide

established global recovery/complementary-channel tools. The latter identifies optimal worst-case entanglement recovery with a complementary-channel optimization. We explicitly use such global dilation alignment in Section 3; our additional input is the finite-word certificate, and the linear diamond estimate uses a subsequent channel-dependent polar calibration rather than just a generic fidelity-to-distance conversion.

Elimelech, Albert and Barg [EAB26, Definition 2 and Theorem 4] construct a single recovery for channels controlled by a prescribed error set [EAB26a]: Kraus operators have a common expansion whose coefficient matrix is a contraction. Their worst-case Bures error is at most the square root of an error-set environment-leakage parameter. Their input is not fixed calibrated output effects. To infer our uniform theorems from that result one would need to construct an error set covering the entire compatible-channel class and bound its leakage from our premise, with the required linear diamond scale. No such conversion is assumed here. We use the term *effects-only uniform decoder* to identify the different public data, not to claim that common recovery maps are new.

Jenčová's channel comparison theorem [J16, Theorem 1] characterizes diamond postprocessing deficiency by guessing experiments on all input ensembles with ancillary systems. Taking the target channel to be the identity gives the global inversion problem. The question here is which finite structured set of binary tests quantitatively suffices; the full family of decision experiments in that characterization is not our premise.

Accordingly, the organizing claims are the matching finite-question modulus, the explicit separation of positive-word cost from recovery cost, and the stated uniform/executable constructions. No claim of a new general error-correction framework or exhaustive priority clearance is intended.

2. Definitions and theorem map

All matrix spaces are finite dimensional. Channels are Schrödinger-picture CPTP maps; their adjoints are UCP. The operator norm is unnormalized, and $\tau_d = \text{Tr} / d$. Let $R = (R_1, \dots, R_g) \subset M_d$ be a finite family of Hermitian unitaries and $F_y = (I + R_y)/2$. Scalar reflections are allowed but carry no information. Trace distance means $\frac{1}{2}\|\rho - \sigma\|_1$. We use the normalized Choi state $J(\Lambda) = (\Lambda \otimes \text{id}_d)(|\Omega_d\rangle\langle\Omega_d|)$, where $|\Omega_d\rangle = d^{-1/2} \sum_i |ii\rangle$; in particular $J(\text{Dep}) = I_{d^2}/d^2$. For a reference register $\mathcal{R}$, write $\Lambda_{\mathcal{R}} = \Lambda \otimes \text{id}_{\mathcal{R}}$. For $E : M_d \rightarrow M_m$, define

$$\epsilon_R(E) = \max_y \min_{0 \leq Q \leq I_m} \|E^*(Q) - F_y\|, \quad \delta_{\text{rec}}(E) = \min_{D: M_m \rightarrow M_d \text{ CPTP}} \frac{1}{2} \|DE - \text{id}_d\|_{\diamond}. \quad (2.1)$$

Compactness supplies the minima. The minimizing effect is allowed to differ for each question. Operator-norm error is the worst error in its acceptance probability over all input states. No joint measurement of the effects is assumed. The recovery error includes arbitrary input-reference entanglement.

The minimax modulus, including arbitrary finite output dimension, is

$$\mathfrak{M}_R(e) = \sup_{m < \infty} \sup_{E: \epsilon_R(E) \leq e} \delta_{\text{rec}}(E). \quad (2.2)$$

For fixed physical effects $Q = (Q_y) \subset M_m$, compatibility means $\max_y \|E^*Q_y - F_y\| \leq e$. A *uniform decoder* is one map $D_{R,Q}$ working for every compatible E ; its construction must not depend on

E . This is stronger than (2.1). A coherent tester is stronger input data still: a specified reversible circuit B_y , with clean workspace initially $|0^a\rangle$, whose acceptance projection induces Q_y .

Theorem 3.1 Premise: Positive word twirl, second moment M_2 , replacer weight λ ; Recovery conclusion: $6M_2e/(1 - \sqrt{1 - \lambda})^2$; Decoder dependence: May depend on E .

Corollary 4.2 Premise: Fixed test count and fixed lazy HS gap; Recovery conclusion: Minimax $\Theta(\min\{1, e \log^2 d\})$; Decoder dependence: As in (2.1).

Theorems 5.1–5.2 Premise: Rank-one anchor and swaps; Recovery conclusion: $6(9d - 8)e$, or uniform $[10(9d - 8) + \sqrt{25d - 24}]e$; Decoder dependence: Respectively E -dependent or effects-only.

Theorem 6.1 Premise: Finite irreducible qubit family; Recovery conclusion: Uniform $1028\Gamma_{\text{cb}}^2 e$; Decoder dependence: Effects-only.

Theorem 7.2 Premise: Seven specified coherent tester types, $n = 3s$; Recovery conclusion: Uniform $4(16n - 14)^2 e$, $32n - 28$ tester calls; Decoder dependence: Tester circuits only.

All displayed error upper bounds are capped by one. None of the mathematical spectral or polar constructions alone carries a gate-efficiency guarantee.

For later comparison, define the complete commutator constant by

$$\Gamma_{\text{cb}}(R) = \sup_{r \geq 1} \sup_X \frac{\|X - I_d \otimes (\tau_d \otimes \text{id}_r)(X)\|}{\max_y \|[X, R_y \otimes I_r]\|}. \quad (2.3)$$

The supremum is over nonzero denominators; if a nonzero numerator can have zero denominator, define the constant as infinity. Ratios 0/0 contribute zero. For scalar common commutant the constant is finite: the finite-dimensional unital $*$ -algebra generated by the reflections is M_d , so express every matrix unit as a finite linear combination of words. Word telescoping bounds each amplified $[X, E_{ij} \otimes I]$ by a finite constant times the denominator in (2.3). Finally,

$$X - I_d \otimes (\tau_d \otimes \text{id})(X) = \frac{1}{d} \sum_{i,j} [X, E_{ij} \otimes I] (E_{ji} \otimes I).$$

The resulting bound is independent of amplification size; no uniform bound across different logical families is asserted by this argument. The centering in (2.3) is the normalized conditional expectation, not a freely optimized scalar. In the pinned version [AJW, Definition 3.1, Eq. (3.3), Eq. (3.18), and Definition 3.8], the corresponding matrix commutator seminorm and complete cost of the fixed-point expectation define the expected length $\kappa(R)$. For scalar commutant this equals (2.3), including its normalization. Indeed the self-adjoint restriction makes no change after amplification: replace an arbitrary X by

$$\mathcal{H}(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}.$$

After reordering tensor factors this doubles only the reference dimension. Centering commutes with this construction, and $\|\mathcal{H}(X)\| = \|X\|$. For each self-adjoint R_y , the commutator has off-diagonal blocks $[X, R_y \otimes I]$ and $-[X, R_y \otimes I]^*$, so it also preserves the commutator norm. Every arbitrary-operator ratio is therefore a self-adjoint ratio at twice the reference size; the reverse inequality is inclusion. No factor-two loss or new metric is asserted.

3. Recovery from a positive word certificate

A formal word $w = y_\ell \cdots y_1$ has length $|w| = \ell$ and unitary $R_w = R_{y_\ell} \cdots R_{y_1}$. For a probability distribution μ with finite second moment, set

$$T_\mu = \sum_w \mu(w) \text{Ad}_{R_w}, \quad M_2 = \sum_w \mu(w) |w|^2, \quad \text{Dep}(X) = \tau_d(X) I_d.$$

The notation $T_\mu \geq_{\text{cp}} \lambda \text{Dep}$ means the difference is completely positive. It is a quantum Doeblin minorization with a specified physical replacer [H24, GHNW26], not a new definition of mixing.

Theorem 3.1. Positive-word recovery

If $0 < \lambda \leq 1$ and $T_\mu \geq_{\text{cp}} \lambda \text{Dep}$, then

$$\delta_{\text{rec}}(E) \leq \min \left\{ 1, \frac{6M_2 \epsilon_R(E)}{(1 - \sqrt{1 - \lambda})^2} \right\}. \quad (3.1)$$

Proof. Write $e = \epsilon_R(E)$, choose the independently calibrated effects, and let $V : \mathbb{C}^d \rightarrow \mathbb{C}^m \otimes \mathcal{F}$ be a Stinespring isometry. On $K = \mathbb{C}^m \oplus \mathbb{C}^m$, use the common projection-dilation space

$$\mathcal{P}(Q) = \begin{pmatrix} Q & \sqrt{Q(I-Q)} \\ \sqrt{Q(I-Q)} & I-Q \end{pmatrix}, \quad T_y = 2\mathcal{P}(Q_y) - I. \quad (3.2)$$

Pad V into the first copy. If $X = V^* \mathcal{P}(Q) V$ approximates a projection P within e , direct multiplication gives

$$(\mathcal{P}(Q)V - VP)^*(\mathcal{P}(Q)V - VP) = (I - P)X(I - P) + P(I - X)P \leq eI. \quad (3.3)$$

Hence $\|T_y V - V R_y\| \leq 2\sqrt{e}$, and word telescoping gives $\|T_w V - V R_w\| \leq 2|w|\sqrt{e}$. Tensor identities on inaccessible registers are suppressed here and below.

Let E^c be complementary to E , let $f(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$, and put $b^2 = 2(1 - f)$. For every state on the input and any reference, the preceding isometries are purifications of $E_{\mathcal{R}}^c \rho$ and $E_{\mathcal{R}}^c \text{Ad}_{R_w, \mathcal{R}} \rho$. Their distance yields $f \geq 1 - 2|w|^2 e$. Joint concavity of root fidelity therefore implies

$$b(E_{\mathcal{R}}^c \rho, E_{\mathcal{R}}^c T_{\mu, \mathcal{R}} \rho) \leq 2\sqrt{M_2 e}. \quad (3.4)$$

Set $C(\rho) = E^c(I/d) \text{Tr} \rho$ and $B = \sup_{\mathcal{R}, \rho} b(E_{\mathcal{R}}^c \rho, C_{\mathcal{R}} \rho)^2$. For $\lambda < 1$, $S = (T_\mu - \lambda \text{Dep})/(1 - \lambda)$ is CPTP, so it preserves the reference marginal. Thus $C_{\mathcal{R}} S_{\mathcal{R}} \rho = C_{\mathcal{R}} \rho$. Convexity of squared Bures distance and the triangle inequality imply

$$\sqrt{B} \leq 2\sqrt{M_2 e} + \sqrt{1 - \lambda} \sqrt{B}, \quad B \leq \frac{4M_2 e}{(1 - \sqrt{1 - \lambda})^2}. \quad (3.5)$$

For $\lambda = 1$ the same inequality follows without a residual channel.

The operational-fidelity dilation theorem [KSW06, Eqs. (25)–(26)] aligns Stinespring representations of E^c and C by one common isometry with squared operator distance at most B . This is a single alignment for all inputs, not state-by-state Uhlmann choices. Applied to the complement, its operation is on the original channel’s accessible output. After finite padding, the alignment acts only on that output and preserves the isometry identity. It supplies a legal decoder isometry whose recovered dilation Z satisfies $\|Z - I \otimes \eta\|^2 \leq B$ for one unit vector η .

To retain a linear bound, isolate the Kraus operator $K_0 = (I \otimes \langle \eta |)Z$. The other Kraus operators have total effect $G = I - K_0^* K_0$, with $\|G\| \leq B$. In a polar decomposition $K_0 = U\sqrt{I - G}$, the partial isometry extends to a unitary, even if singular. Correct it on the output. The remaining channel is $X \mapsto \sqrt{I - G}X\sqrt{I - G} + \Psi(X)$, with Ψ CP and $\Psi^*I = G$. Since

$$\|\Psi\|_\diamond = \|G\|, \quad \|\text{Ad}_{\sqrt{I - G}} - \text{id}\|_\diamond \leq 2\|I - \sqrt{I - G}\| \leq 2\|G\|,$$

the half-diamond error is at most $3B/2$. Substitution proves (3.1). The final unitary may depend on E . No environmental access is used. $\square$

The common direct-sum dilation in (3.2) uses one clean qubit, not a fresh ancilla for each word letter. Subsequent dilation alignment may require additional finite private output space. This theorem proves existence of a CPTP decoder, not an efficient implementation or a closed reusable apparatus.

Corollary 3.2. A lazy Hilbert–Schmidt gap

Let $L = \frac{1}{2}\text{id} + \frac{1}{2g} \sum_y \text{Ad}_{R_y}$. Suppose $\|L(X)\|_2 \leq (1 - \Delta)\|X\|_2$ for every traceless X , where $0 < \Delta < 1$. Set

$$k = \left\lceil \frac{\ln(4d^2)}{-\ln(1 - \Delta)} \right\rceil.$$

Then $\delta_{\text{rec}}(E) \leq \min\{1, 6(k^2 + k)\epsilon_R(E)\}$.

Proof. For the normalized Choi state and $|\Omega\rangle = d^{-1/2} \sum_i |ii\rangle$, HS contraction on the first traceless factor gives

$$\|J(L^k) - I/d^2\|_{\text{op}} \leq \|J(L^k) - I/d^2\|_2 \leq (1 - \Delta)^k \leq 1/(4d^2).$$

Thus $L^k \geq_{\text{cp}} (3/4)\text{Dep}$. Its syntactic word length is $N \sim \text{Binomial}(k, 1/2)$, so $M_2 = (k^2 + k)/4$. Apply Theorem 3.1. If $\Delta = 1$, then $L = \text{Dep}$ and its one-step second moment $1/2$ gives the stronger bound $3e$. $\square$

Only constant replacer weight is required; mixing need not continue to accuracy e . This explains the absence of $\log(1/e)$.

4. A matching word-filtration obstruction

Theorem 4.1. Word-ball lower bound

Let $\ell \geq 1$ be an integer. If $\sum_{j=0}^{\ell-1} g^j < d$, then for every $e \geq 0$,

$$\mathfrak{M}_R(e) \geq \min \left\{ \frac{1}{2}, \frac{e}{1 - \cos(\pi/(2\ell))} \right\} \geq \min \left\{ \frac{1}{2}, \frac{8}{\pi^2} \ell^2 e \right\}. \quad (4.1)$$

Proof. Fix a unit vector u , let $K_j = \text{span}\{R_w u : |w| \leq j\}$, and let P_j project onto K_j . Then $K_{\ell-1} \neq \mathbb{C}^d$. Define $D = \sum_{j=0}^{\ell-1} (I - P_j)$. Its orthogonal filtration levels $H_0 = K_0$, $H_j = K_j \ominus K_{j-1}$, and $H_\ell = K_{\ell-1}^\perp$ have eigenvalues $0, 1, \dots, \ell$. Some intermediate levels may be zero. Every R_y is banded to adjacent levels, because $R_y K_j \subset K_{j+1}$ and $R_y = R_y^*$.

Put $S = e^{i\pi D}$, $B_y = (R_y - S R_y S)/2$, so $\|B_y\| \leq 1$. For $\theta = \pi/(2\ell)$, the channel

$$E_0 = \frac{1}{2} \text{Ad}_{e^{i\theta D}} + \frac{1}{2} \text{Ad}_{e^{-i\theta D}}$$

fixes diagonal filtration blocks and multiplies adjacent blocks by $\cos \theta$. Taking $Q_y = F_y$ gives $\epsilon_R(E_0) \leq a = (1 - \cos \theta)/2$.

Choose a unit $v \in H_\ell$. The orthogonal states $\rho_\pm = |(u \pm v)/\sqrt{2}\rangle\langle(u \pm v)/\sqrt{2}|$ have identical images under E_0 . Under $E_p = (1 - p)\text{id} + pE_0$ their trace distance is $1 - p$. If a decoder has half-diamond error δ , triangle inequality and contractivity give $1 \leq 2\delta + 1 - p$; hence $\delta \geq p/2$. Choose $p = \min(1, e/a)$. Since $\epsilon_R(E_p) \leq pa$, the resulting channel is admissible for (2.2) and yields (4.1). The zero-error case follows by taking $p = 0$. $\square$

This is a minimax lower construction. Its optimized question error is only bounded above by the selected budget; equality with that budget is unnecessary.

Corollary 4.2. Fixed-degree, fixed-gap sharpness

Along any family with fixed $g \geq 2$, lazy HS gap at least $\Delta_0 > 0$, and dimensions tending to infinity,

$$\mathfrak{M}_R(e) = \Theta_{g, \Delta_0}(\min\{1, e(\log d)^2\}). \quad (4.2)$$

Indeed $\ell = \lfloor \log_g((g-1)d) \rfloor$ satisfies the strict word-ball inequality, and $\ell = \Theta_g(\log d)$. Corollary 3.2 gives the upper bound uniformly in all finite output dimensions.

Constant-degree quantum expanders have an earlier constructive literature, including Ben-Aroya, Schwartz and Ta-Shma [BST10]. For the explicit reflection-only family used here, Anshu, Bala-subramanian, Haferkamp, Harrow and Tan [ABHHT26, Theorem 2.2] give sixteen Pauli/CNOT involutions on $n = 3s$ qubits with a dimension-independent HS gap. Their non-lazy gap is at least $1/(2 \cdot 10^8)$; lazification gives at least $1/(4 \cdot 10^8)$. This supplies an external expander input, not a new expander construction here. This realization is pinned to the September 2026 v1 statement cited below. The conditional modulus (4.2), the signed-family results, and the explicit circuit identities in Section 7 do not depend on that preprint's gap proof.

5. Direct extraction can outperform positive averaging

Let $P = |1\rangle\langle 1|$, $Z_1 = I - 2P$, and let T_{ij} swap coordinates i, j . The anchor tests are $Z_1, T_{12}, \dots, T_{1d}$. For the geometry comparison we use the larger family $\mathcal{R}_d = \{Z_1\} \cup \{T_{ij} : i < j\}$, with $d \geq 3$.

Theorem 5.1. Channel-dependent anchor recovery

The anchor tests alone imply $\delta_{\text{rec}}(E) \leq \min\{1, 6(9d - 8)\epsilon_R(E)\}$.

Proof. Complement the effect for Z_1 to obtain a physical effect Q_0 approximating P . Dilate all effects on the same $K = \mathbb{C}^{2m}$; write $B = \mathcal{P}(Q_0)$, $U_i = 2\mathcal{P}(Q_i) - I$ for $i \geq 2$, and $U_1 = T_1 = I$, $T_i = T_{1i}$. Let V denote the padded Stinespring isometry and set $\eta = V|1\rangle$, $J\psi = \psi \otimes \eta$. Define, on the accessible output alone,

$$A : K \rightarrow \mathbb{C}^d \otimes K, \quad Ax = \sum_{i=1}^d |i\rangle \otimes BU_i x. \quad (5.1)$$

Equation (3.3) gives $\|BV - VP\| \leq \sqrt{e}$ and $\|U_i V - VT_i\| \leq 2\sqrt{e}$. Since $PT_i = |1\rangle\langle i|$, the row errors of $AV - J$ are at most $\sqrt{e}, 3\sqrt{e}, \dots$. The column errors of $A^*J - V$ have the same bounds, by applying the intertwiner estimates to $|1\rangle$. Thus, with $r^2 = (9d - 8)e$,

$$\|AV - J\|, \quad \|A^*J - V\| \leq r. \quad (5.2)$$

Extend the polar partial isometry to an isometry W , so $A = WS$, $S \geq 0$. For arbitrary v, j , write $z = W^*j$ and $j_\perp = j - Wz$. Then

$$\begin{aligned} \|Av - j\|^2 + \|A^*j - v\|^2 &= \frac{1}{2}\|(S - I)(v + z)\|^2 + \frac{1}{2}\|(S + I)(v - z)\|^2 + \|j_\perp\|^2 \\ &\geq \frac{1}{2}\|Wv - j\|^2. \end{aligned} \quad (5.3)$$

The inequality persists after tensoring with an identity. Hence $\|WV - J\|^2 \leq 4r^2$. The decoder applying W to the padded output and discarding K is physical; the polar calibration in the last paragraph of Theorem 3.1 supplies a further output unitary and error at most $\frac{3}{2} \cdot 4r^2 = 6r^2$. Singular values of A need not be bounded away from zero. The further unitary may depend on E . $\square$

Here the direct sum $K = \mathbb{C}^{2m}$ is implemented by one initially clean qubit. The extraction isometry maps K into $\mathbb{C}^d \otimes K$, and its multiplicity register is discarded. Private workspace and disposal are permitted; neither cost nor clean return of that register is promised by this mathematical existence theorem.

Theorem 5.2. One anchor decoder from the effects alone

Fix the physical effects. The same isometric polar extension W of (5.1), without a channel-dependent final correction, defines a decoder D_0 with

$$\sup_{E \text{ compatible}} \frac{1}{2}\|D_0 E - \text{id}\|_\diamond \leq \min\{1, [10(9d - 8) + \sqrt{25d - 24}]e\}. \quad (5.4)$$

Proof. Fix E only for analysis. Let $Z = W^*J$, $R_1 = SV - Z$, $R_2 = SZ - V$. Equation (5.2) gives both residual norms at most r . As $(S + I)(V - Z) = R_1 - R_2$, $\|V - Z\| \leq 2r$ and $\|(I - S)V\| \leq 3r$. Equation (5.3) already gives $\|WV - J\| \leq 2r$.

Put $F_0 = J^*AV$, $K_0 = J^*WV$. The identity

$$K_0 - F_0 = V^*(I - S)V + (Z - V)^*(I - S)V \quad (5.5)$$

separates a Hermitian term and an error of norm at most $6r^2$. Also $I - \text{Re } K_0 = (WV - J)^*(WV - J)/2$, of norm at most $2r^2$. Taking real and imaginary parts in (5.5) gives $\|K_0 - I\| \leq \|F_0 - I\| + 8r^2$. The leakage effect away from the environmental vector η has norm $\|I - K_0^*K_0\| \leq 4r^2$. Consequently

$$\frac{1}{2}\|D_0E - \text{id}\|_\diamond \leq \|F_0 - I\| + 10r^2. \quad (5.6)$$

It remains to use compression rather than an amplitude estimate for F_0 . Write $X = V^*BV$, $Y_i = V^*U_iV$. Then

$$V^*BU_iV = XY_i + V^*B(I - VV^*)U_iV.$$

The first term is within $3e$ of PT_i . The second has norm at most $\sqrt{\|X - X^2\|}\sqrt{\|I - Y_i^2\|} \leq 2e$. Here $0 \leq X \leq I$ is within e of a projection, giving $\|X - X^2\| \leq e$, and the reflection compression has defect at most $4e$. For the first assertion, when $e < 1/4$ the spectrum of X lies within e of $\{0, 1\}$, so every eigenvalue x satisfies $x(1 - x) \leq e$; for larger e , the universal bound $1/4$ suffices. The covariance factorization follows by inserting $I - VV^*$: its two squared factor norms are respectively $\|V^*B(I - VV^*)BV\| = \|X - X^2\|$ and $\|V^*U_i(I - VV^*)U_iV\| = \|I - Y_i^2\|$. The first row, $i = 1$, costs just e . Since $F_0 = \sum_i |i\rangle\langle 1|V^*BU_iV$, row summation gives $\|F_0 - I\| \leq \sqrt{25d - 24}e$. Apply (5.6). The choice of W was made before E ; its arbitrary kernel completion is fixed throughout. $\square$

Proposition 5.3. Signed-permutation lower bound and geometry

For $\mathcal{R}_d$,

$$c_0 \min\{1, de\} \leq \mathfrak{M}_{\mathcal{R}_d}(e) \leq \min\{1, 54de\}, \quad c_0 = (1 - 1/\sqrt{2})/2. \quad (5.7)$$

Moreover $\Gamma_{\text{cb}}(\mathcal{R}_d) = \Theta(\sqrt{d})$, words of length at most four span M_d , but every positive-word coefficient in (3.1) is at least $\frac{3}{2}(d - 1)^2$.

Proof of the lower bound. Put $u = d^{-1/2}(1, \dots, 1)$, $S = I - 2|u\rangle\langle u|$, and $E_p = (1 - p)\text{id} + p\text{Ad}_S$, $0 \leq p < 1/2$. Every swap commutes with S . For $v \perp u$, the two states $|u \pm v\rangle/\sqrt{2}$ contract in trace distance by $1 - 2p$, so every recovery has error at least p . The identity decoder attains p ; hence $\delta_{\text{rec}}(E_p) = p$.

In the plane containing u and e_1 , the rank-one anchor has Bloch coordinates $(s_d, 0, -t_d)$, where $s_d = 2\sqrt{d-1}/d$, $t_d = (d-2)/d$, and $s_d^2 + t_d^2 = 1$. The channel shrinks its transverse coordinate by $c = 1 - 2p$. Choose as output effect the projection with Bloch vector $\beta(s_d/c, 0, -t_d)$, where $\beta = (t_d^2 + s_d^2/c^2)^{-1/2}$, and extend it by zero on the orthogonal complement. Its pullback differs from the anchor by $e_d(p) = (1 - \beta)/2$. Complement it for the positive Z_1 test. Thus $\epsilon_{\mathcal{R}_d}(E_p) \leq e_d(p)$. Solving $e = e_d(p)$, for $e < 1/2$, gives

$$\mathfrak{M}_{\mathcal{R}_d}(e) \geq f\left(\frac{(1-2e)^{-2}-1}{s_d^2}\right), \quad f(x) = \frac{1}{2}(1-(1+x)^{-1/2}).$$

The argument is at least de ; monotonicity and concavity give $f(de) \geq f(1) \min(1, de)$. For $e \geq 1/2$, use monotonicity of the modulus. The upper bound is Theorem 5.1. $\square$

Proof of the geometric comparison. At any amplification let $h = \max_R \| [X, R \otimes I] \|$. Because $Z_i = T_{1i} Z_1 T_{1i}$, each off-diagonal block row of X has norm at most $3h/2$. Its row domain is the complement of that coordinate, so orthogonal-range summation bounds the off-diagonal part by $\frac{3}{2}\sqrt{d-1}h$. The (i, j) block of $[X, T_{ij}]$ is $X_{ii} - X_{jj}$; the diagonal part is within $(d-1)h/d$ of its normalized partial trace. Thus

$$\Gamma_{\text{cb}} \leq \frac{3}{2}\sqrt{d-1} + (d-1)/d.$$

Testing (2.3) on $|u\rangle\langle u|$ gives the lower bound $\sqrt{d-1}/2$. Algebraic span in radius four follows from $E_{ii} = (I - Z_i)/2$, $T_{ij} - I + E_{ii} + E_{jj} = E_{ij} + E_{ji}$, and $Z_i T_{ij} - I + E_{ii} + E_{jj} = E_{ji} - E_{ij}$.

Every length- ℓ word is a signed permutation moving at most 2ℓ coordinates. Let $\Pi_{\text{off}} = \sum_{i \neq j} |ij\rangle\langle ij|$. Its expectation in $(R_w \otimes I)|\Omega\rangle$ is at most $2|w|/d$. Choi domination of T_μ by λDep therefore forces $\mathbb{E}|w| \geq \lambda(d-1)/2$, and $M_2 \geq \lambda^2(d-1)^2/4$. Since $\lambda/(1-\sqrt{1-\lambda}) = 1+\sqrt{1-\lambda}$, the coefficient in (3.1) is at least $3(d-1)^2/2$. Conversely every signed permutation has length at most $4d-3$, and uniform averaging over them is exactly depolarizing. Constant-weight minorization radius is therefore $\Theta(d)$. $\square$

The last statement limits this particular positive-averaging certificate, not physical recovery itself. For the established comparison of cost-induced mixing, cb mixing and CP return times, see [AJW, Section 3.3, Eq. (3.50)]. That hierarchy does not itself state our signed-family recovery separation.

6. Uniform recovery of an arbitrary logical qubit

Theorem 6.1. Fixed-effect qubit recovery

For a finite qubit reflection family with scalar common commutant and fixed effects $Q_y \in M_m$, there is one decoder depending only on R, Q such that every compatible channel satisfies

$$\frac{1}{2} \|D_{R,Q}E - \text{id}_2\|_\diamond \leq \min\{1, 1028\Gamma_{\text{cb}}(R)^2 e\}. \quad (6.1)$$

The decoder adds a factor-four clean space, extracts an exact Pauli algebra, and discards its multiplicity. It requires spectral preprocessing, not a promised efficient circuit.

The full proof is Appendix A, included in the complete review artifact. Choose a pair of Bloch axes with maximal sine $s > 0$, rotate logical coordinates so they are Z and $cZ + sX$, and dilate their effects to two physical reflections. A further doubling allows a kernel-completed sign of their commutator that anticommutes exactly with the first reflection. The same physical representation then works for every compatible pullback. Two-sided CP defect estimates control the sign's off-diagonal entry at order e/s^2 , rather than using a generic square-root continuity estimate. The three Pauli bounds imply half-diamond error below $257e/s^2$, and $\Gamma_{\text{cb}} \geq 1/(2s)$.

Extraction from supplied measurements is established in complementary-observable recovery [R16,NMK25] and self-testing [SH16,SBWS18]. The commutator spectral pairing used to identify a Pauli factor also appears in [K17, Proposition A.3]. Appendix A uses this representation mechanism and standard CP covariance estimates [C74] to bound the recovered Pauli pullbacks linearly in operator norm, hence in half-diamond distance. The quantitative conclusion includes possibly oblique calibrated pairs; the extraction principle itself is prior. The square-root obstruction in [SH16, Section III.3] additionally requires post-measurement certification and does not obstruct the recovery channel here.

7. An executable decoder from seven coherent test types

Lemma 7.1. Extracting an output while discarding leakage

Let $J = J_D \otimes I_O$ be an isometry and $P = JJ^* = P_D \otimes I_O$. Compare an ideal unitary circuit W with a physical unitary circuit $\widetilde{W}$, containing L approximate steps and exact interspersed gates that intertwine J . Assume at step t

$$\|J^*\widetilde{G}_tJ - G_t\| \leq \delta_t, \quad \|(I - P)\widetilde{G}_tJ\| \leq \lambda_t^{\text{out}}, \quad \|J^*\widetilde{G}_t(I - P)\| \leq \lambda_t^{\text{in}}.$$

If the ideal circuit extracts the desired output into O , the physical one followed by discarding all of D has half-diamond distance from the ideal retained-output channel at most

$$\sum_t \delta_t + \sum_t \lambda_t^{\text{in}} \sum_{i < t} \lambda_i^{\text{out}} + \frac{1}{2} \left(\sum_t \lambda_t^{\text{out}} \right)^2. \quad (7.1)$$

Proof. For the prefix let $q_t = \|(I - P)\widetilde{W}_tJ\|$. Inserting $P + (I - P)$ gives $q_t \leq q_{t-1} + \lambda_t^{\text{out}}$. Likewise, for $a_t = \|J^*\widetilde{W}_tJ - W_t\|$, $a_t \leq a_{t-1} + \delta_t + \lambda_t^{\text{in}}q_{t-1}$. Thus a_L is bounded by the first two terms of (7.1).

After initializing O , decompose the final isometry as $JK + Z$, with $PZ = 0$. Because $P = P_D \otimes I_O$, all compressed/leakage cross terms vanish on tracing D , including with a reference: every retained observable commutes with P . The compressed CP contribution differs from the ideal one in diamond norm by at most $2a_L$; the leakage CP contribution has diamond norm $\|Z\|^2 \leq q_L^2$. Divide by two. $\square$

There is no physical projection onto P . Retaining any part of its support instead of discarding it can invalidate the cross-term cancellation.

Theorem 7.2. Cyclic coherent-tester extraction

Let $n = 3s$, and partition logical computational labels into $(u, v, w) \in (\mathbb{F}_2^s)^3$. Let the seven test reflections be $X_1, Z_1, C_{12}, C_{21}, C_{23}, C_{32}, R$, where C_{ab} applies a CNOT from coordinate a to b in every triple and R sends (u, v, w) to $(u, v + Su, w)$, with cyclic shift $(Su)_i = u_{i+1 \bmod s}$. These are among the explicit generators of [ABHHT26, Theorem 2.2].

Suppose each output test is supplied by a reversible circuit B_y of at most G elementary gates, padded to a common a -qubit workspace, and acceptance is flag value one. There is one decoder $D_{\mathbf{B}}$, determined by the supplied tester circuits alone, such that every compatible channel satisfies

$$\frac{1}{2}\|D_{\mathbf{B}}E - \text{id}_{2^n}\|_{\diamond} \leq \min\{1, 4(16n - 14)^2 e\}. \quad (7.2)$$

It uses $32n - 28$ forward/inverse tester calls, $n + a$ initially clean qubits, $O(nG + n^2)$ ordinary gates and $O(n^3 + nG)$ classical compilation work (apart from representation/addressing costs). It makes no additional calls to E , uses no postselection, and retains only the new output register. No expander gap is required for this construction.

Proof. Put $S_{ab} = C_{ab}C_{ba}C_{ab}$, so these swap coordinate blocks. The seven-letter word $U = RS_{12}S_{23}$, rightmost first, acts by $(u, v, w) \mapsto (w, u + Sw, v)$, with binary matrix

$$A = \begin{pmatrix} 0 & 0 & I \\ I & 0 & S \\ 0 & I & 0 \end{pmatrix}.$$

For $\bar{S} = \text{diag}(S, S, S)$, direct multiplication gives $A^3 + \bar{S}A + I = 0$, hence $\bar{S} = A^2 + A^{-1}$. Let f be the first coordinate of the u block. Its forward cyclic space is invariant under A^{-1} , since A is invertible on that finite-dimensional invariant space. It therefore contains all shifts of f, Af, A^2f , which are the first coordinates of u, v, w . Hence $f, Af, \dots, A^{n-1}f$ form a basis. Transposing the recurrence proves the same statement for A^T : its first three powers on f occupy the first coordinates of u, w, v , and its cyclic space is invariant under $\bar{S}^T = (A^T)^2 + (A^T)^{-1}$. The shifts fill all coordinates. In either case, if the first n iterates were dependent, the first such dependence would make the preceding span invariant and prevent it from reaching dimension n . Set

$$F = A^{n-1}, \quad N = \begin{pmatrix} f^T \\ f^T A \\ \vdots \\ f^T A^{n-1} \end{pmatrix}, \quad M = [A^{-(n-1)}f \ A^{-(n-2)}f \ \dots \ f]. \quad (7.3)$$

These are controllability and observability matrices over $\mathbb{F}_2$. Both M, N are invertible. The following record-then-erase transfer has the same basic pattern as CSS-style coherent decoding [NMK25, Section III.1]. Append output $O = |0^n\rangle$. Apply Hadamards to O , then for $t = 0, \dots, n-1$ apply Z_1 controlled by O_t , followed by U unless $t = n-1$. Final Hadamards give exactly $|x\rangle|0\rangle \mapsto |Fx\rangle|Nx\rangle$, because the phase on label b is $(-1)^{b^T Nx}$.

Apply $M^{-1}N^{-1}$ to O , obtaining label $c = M^{-1}x$. Next apply X_1 controlled by O_t , followed by U^{-1} unless $t = n-1$. The final old logical label is $A^{-(n-1)}Fx + \sum_t c_t A^{-(n-1-t)}f = x + Mc = 0$. Apply M to O . Thus the ideal circuit transfers every quantum state coherently, with $W(|\psi\rangle|0\rangle) = |0\rangle|\psi\rangle$. It has $L = 2n + 14(n-1) = 16n - 14$ reflection occurrences.

Replace each reflection by $\hat{R}_y = B_y^*(2\Pi_y - I)B_y$. A controlled version uses the unconditional B_y , controlled middle phase, and unconditional B_y^* . For acceptance flag one that middle phase is $Z_{\text{control}}CZ_{\text{control,flag}}$; omitting the first sign would implement the wrong controlled reflection.

For analysis alone choose the padded Stinespring isometry V_0 for E , including the tester workspace initially zero, and set $J = V_0 \otimes I_O$. Each reflection's compressed discrepancy is $\delta_t \leq 2e$. Hermiticity and the identity

$$(\tilde{G}J - JG)^*(\tilde{G}J - JG) = 2I - (J^*\tilde{G}J)G - G(J^*\tilde{G}J)$$

give both leakage bounds at most $2\sqrt{e}$. Controlled reflections have the same blockwise estimates. Lemma 7.1 gives $2Le + 4e(L^2 - L/2) = 4L^2e$.

The common tester workspace may be dirty between occurrences. Its arbitrary off-subspace action is already covered by the lemma; no fresh promise or reset is assumed. All noisy-output and workspace registers are finally discarded, as is the inaccessible environment in the channel description. The decoder never uses that environment or the analytical projector.

Each reflection uses two tester calls. Binary Gaussian elimination compiles the two output linear maps with $O(n^2)$ CNOT/SWAP gates and $O(n^3)$ classical work. Copying/reversing tester descriptions costs $O(nG)$. This proves the resources and uniformity. $\square$

There are two distinct resource accounts. In the reversible black-box model, count $32n - 28$ calls to testers or their inverses and $O(n^2)$ additional Clifford gates; a controlled tester call is not assumed. In the supplied circuit-description model, expand those calls to obtain total gate count $O(nG + n^2)$ and copying/compilation work $O(n^3 + nG)$. The input channel's output register is supplied, not counted among the $n + a$ newly clean qubits.

If conversion to another gate alphabet requires approximation, synthesizing each of N_g gates to operator error η/N_g adds at most η to the half-diamond error by telescoping; use the actual gate model's synthesis cost rather than identifying exact spectral preprocessing with an efficient algorithm. With the supplied exact tester circuits and Clifford additions, no new approximation is required.

The seven-type family is itself irreducible: its conjugated X_1, Z_1 directions span all binary Pauli coordinates by (7.3), and hence generate the full matrix algebra. Theorem 4.1 supplies an $\Omega(n^2e)$ minimax obstruction for this fixed test count. Theorem 7.2 matches this error order under coherent tester access. It does not bound the cost of obtaining those testers from an unknown channel or from classical target advice.

8. Interpretation and limitations

Generation is not a quantitative certificate. The signed-permutation family has algebraic span in radius four, yet its question error can be a factor of order d smaller than its recovery error. Positive mixing and direct extraction can have different costs even for that same family.

Uniform recovery requires a fixed construction. Optimizing a decoder after knowing E does not justify swapping its quantifier with a family of channels. The uniform anchor and qubit constructions avoid that issue by fixing their operators from the effects before choosing E for analysis.

Sharp effects are structural input. They supply two small CP covariance factors, so multiplying two calibrated sharp representatives can cost $O(e)$, not automatically $O(\sqrt{e})$. This local fact is not a dimension-free bound on arbitrarily many products or on a global UCP section.

Unused physical sectors must remain harmless. Append an unused two-dimensional sector to an exact four-dimensional encoding. Physical test operators can have incompatible behavior on that sector while all pulled-back logical tests and recovery remain exact. Thus a necessary recovery geometry cannot demand small global physical commutators on irrelevant output space.

The unresolved intrinsic question. For general finite irreducible reflection families, it remains open here whether $\delta_{\text{rec}}(E) \leq CT_{\text{cb}}(R)^2 \epsilon_R(E)$ with an absolute constant. The uniform version,

requiring one decoder from fixed effects, is a distinct stronger target. The qubit and signed-permutation results do not prove either claim in arbitrary dimension.

Operational exclusions. No result promises reusable or clean garbage, an autonomous repeated-use apparatus, or a protocol deriving the calibration premise from finitely many samples. The operator-norm premise is a mathematical worst-input guarantee. The executable theorem assumes reversible coherent tester access. It therefore supplies no unrestricted Aaronson–Kuperberg unitary-synthesis result without a separate lawful tester construction.

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The arguments are analytic, model-assisted proofs. Supplementary Lean code checks the passage from independently calibrated physical projections to finite-word operator-norm stability on arbitrary complex Hilbert spaces. This is partial formalization of an input to Section 3: dilation of arbitrary effects, channels, fidelity, diamond norm, and the recovery conclusions are not formalized. The remaining local checks concern exact finite controls, typeset preservation and provenance. No complete proof-assistant verification of this paper is claimed.

Appendix A. A decoder fixed by two oblique qubit effects

This appendix proves Theorem 6.1 of the main manuscript. All tensor identities on Stinespring environments are suppressed when no confusion is possible. No result in this appendix assumes access to that environment.

A.1. Fix the decoder before choosing a channel

Discard scalar logical reflections. Write each remaining one as $R_y = n_y \cdot \sigma$. The finite irreducible family has a pair maximizing $s = |n_1 \times n_2| > 0$. A fixed logical unitary and orientation put this pair in the form

$$R_1 = Z, \quad R_2 = cZ + sX, \quad c^2 + s^2 = 1. \quad (\text{A.1})$$

For the two fixed physical effects put $C_i = 2Q_i - I$, and form the Halmos reflection dilations on $\mathbb{C}^{2m}$:

$$A = \begin{pmatrix} C_1 & \sqrt{I - C_1^2} \\ \sqrt{I - C_1^2} & -C_1 \end{pmatrix}, \quad B = \begin{pmatrix} C_2 & \sqrt{I - C_2^2} \\ \sqrt{I - C_2^2} & -C_2 \end{pmatrix}. \quad (\text{A.2})$$

Both square to identity. Double once more: $A' = A \oplus (-A)$, $B' = B \oplus (-B)$, and $H' = (A'B' - B'A')/(2i) = H \oplus H$. Since $A'H'A' = -H'$, the first reflection exchanges the positive and negative spectral spaces of H' . On the nonzero spectrum choose the usual sign. On the remaining space $\ker H \oplus \ker H$, choose instead the copy-exchange map $(\xi, \zeta) \mapsto (\zeta, \xi)$. The resulting operator S obeys

$$S = S^*, \quad S^2 = I, \quad A'S = -SA', \quad H' = S|H'|. \quad (\text{A.3})$$

The exchange rule is essential: arbitrary independent signs on zero modes need not anticommute with A' .

Two anticommuting reflections give an exact qubit representation. More explicitly, S maps the positive eigenspace of A' isometrically onto the negative one, so their dimensions agree. Choose bases related by this map and adjust a phase. This produces a unitary $W : \mathbb{C}^{4m} \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^{2m}$ satisfying

$$WA'W^* = Z \otimes I, \quad WSW^* = Y \otimes I, \quad W(iA'S)W^* = X \otimes I. \quad (\text{A.4})$$

Let $J : \mathbb{C}^m \rightarrow \mathbb{C}^{4m}$ embed into the first copy, and define

$$D_{R,Q}(\rho) = \text{Tr}_{2m}(WJ\rho J^*W^*). \quad (\text{A.5})$$

Transform back by the fixed logical coordinate choice when necessary. This is CPTP and depends only on the logical family and its fixed physical effects. It has been defined before any noise channel, error parameter or Stinespring representation is chosen.

A.2. First and second moments under a compatible pullback

Now fix an arbitrary compatible E . Let V be its Stinespring isometry followed by J , and write $\phi(T) = V^*(T \otimes I)V$. Then ϕ is UCP and

$$a = \phi(A'), \quad b = \phi(B'), \quad \|a - Z\|, \|b - (cZ + sX)\| \leq 2e. \quad (\text{A.6})$$

For UCP maps the standard CP Schwarz/covariance estimate [C74] gives

$$\|\phi(XY) - \phi(X)\phi(Y)\| \leq \|\phi(XX^*) - \phi(X)\phi(X)^*\|^{1/2} \|\phi(Y^*Y) - \phi(Y)^*\phi(Y)\|^{1/2}. \quad (\text{A.7})$$

Indeed insert $I - VV^*$ between X, Y and use the two factor norms. Since A', B' are reflections, their left and right defects coincide, and $\|I - a^2\|, \|I - b^2\| \leq 4e$.

Put $U = A'B'$ and $U_0 = Z(cZ + sX) = cI + isY$. Equation (A.7) and the two calibration errors yield $\|\phi(U) - U_0\| \leq 8e$. Both unitary defects of $u = \phi(U)$ are at most $16e$. Applying (A.7) to U^2 , then comparing u^2 with U_0^2 , gives

$$\|\phi(U^2) - U_0^2\| \leq 32e, \quad \|\phi(U^{*2}) - U_0^{*2}\| \leq 32e. \quad (\text{A.8})$$

As $H' = (U - U^*)/(2i)$ and $H'^2 = (2I - U^2 - U^{*2})/4$,

$$\|h - sY\| \leq 8e, \quad \|k - s^2I\| \leq 16e, \quad h = \phi(H'), \quad k = \phi(H'^2). \quad (\text{A.9})$$

Consequently

$$\|H'V - sVY\|^2 = \|k - shY - sYh + s^2I\| \leq 32e =: q. \quad (\text{A.10})$$

The completed sign S commutes with $|H'|$ and preserves its kernel. These estimates do not assume invertibility or a gap in the physical commutator spectrum. The logical separation $s > 0$ is the only gap used.

A.3. Sign rounding at linear error

Let $\Pi_{\pm} = (I \pm S)/2$, $G = |H'|$, and let $v_{\pm} = V|y_{\pm}\rangle$, where $Y|y_{\pm}\rangle = \pm|y_{\pm}\rangle$. On the wrong sign sector, $(H' - s)\Pi_-v_+ = -(G + s)\Pi_-v_+$. Thus (A.10) gives

$$\|\Pi_-v_+\|^2, \quad \|\Pi_+v_-\|^2 \leq q/s^2. \quad (\text{A.11})$$

This also handles the kernel, where the separation from the logical target eigenvalue is exactly s . The two diagonal entries of $\phi(S) - Y$ have modulus at most $2q/s^2 = 64e/s^2$.

For its off-diagonal entry $t = \langle v_+, Sv_- \rangle$, use

$$\begin{aligned} \langle v_+, H'v_- \rangle - st &= \langle \Pi_+v_+, (G - s)\Pi_+v_- \rangle \\ &\quad - \langle \Pi_-v_+, (G - s)\Pi_-v_- \rangle. \end{aligned} \quad (\text{A.12})$$

In the first term, move the self-adjoint $G - s$ onto $\Pi_+ v_+$, whose residual norm is at most $\sqrt{q}$; its other factor has norm at most $\sqrt{q}/s$. In the second term the residual acts on $\Pi_- v_-$, again a proper-sign factor of norm at most $\sqrt{q}$, and the remaining wrong-sign factor costs $\sqrt{q}/s$. Thus the right side of (A.12) has modulus at most $2q/s$. Equation (A.9) also gives $|\langle v_+, H' v_- \rangle| \leq 8e$. Therefore

$$|t| \leq 8e/s + 2q/s^2 \leq 72e/s^2. \quad (\text{A.13})$$

Row-sum bounding the resulting Hermitian two-by-two matrix yields $\|\phi(S) - Y\| \leq 136e/s^2$. For $g = \phi(S)$, $\|I - g^2\| \leq 272e/s^2$. Applying (A.7) to $A'S$ bounds its covariance by $\sqrt{4e \cdot 272e/s^2} < 33e/s$. Since $iZY = X$,

$$\|\phi(iA'S) - X\| \leq 2e + 136e/s^2 + 33e/s \leq 171e/s^2. \quad (\text{A.14})$$

All three recovered Pauli errors are thus at most $171e/s^2$.

A.4. Complete norm and geometry

For every amplification, write $T = I \otimes T_0 + X \otimes T_x + Y \otimes T_y + Z \otimes T_z$. Each coefficient is obtained by applying the norm-one functional $A \mapsto \tau_2(\sigma_j A)$ on the first factor; its completely bounded norm is one. Hence $\|T_j\| \leq \|T\|$. The adjoint of $D_{R,Q}E - \text{id}$ vanishes on I , so

$$\|D_{R,Q}E - \text{id}\|_\diamond = \|(D_{R,Q}E - \text{id})^*\|_{\text{cb}} \leq 3 \cdot 171e/s^2. \quad (\text{A.15})$$

The half-diamond bound is below $257e/s^2$. If R_i is one member of the selected maximizing pair, then $\|R_i - \tau_2(R_i)I\| = 1$ and $\max_y \|[R_i, R_y]\| \leq 2s$. Therefore $\Gamma_{\text{cb}}(R) \geq 1/(2s)$, giving (6.1).

For each compatible E , the proof chooses a possibly different Stinespring witness but never changes (A.5). Thus the same decoder works uniformly without a convexity assumption on the compatible family. The cap by one treats all larger-error cases. Finiteness of the logical question family ensures the maximizing pair exists.

A.5. Resources and scope

The construction embeds the physical output of dimension m into dimension $4m$, equivalent to adding two clean qubits, applies a fixed unitary identification and discards the multiplicity of dimension $2m$. There is no postselection, environmental access, extra channel use, or channel-dependent correction. Exact square roots, a spectral sign, a kernel split and the unitary identification are mathematical preprocessing. No polynomial-time synthesis claim follows from their existence.